Supplementary Information

Evaluation Choices Determine Apparent Quantum-Kernel Robustness under Distribution Shift

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Supplementary Note: Scope

This document provides the construction details, complete candidate definitions, secondary analyses, and reporting specifications that support the main article. The primary estimand and conclusions are unchanged: the eight scenario-groups are fixed case studies; the headline comparison selects configurations without out-of-distribution (OOD) labels, tunes regularization from training data only, and gives the two extended kernel families equal candidate budgets. All tables and figures are generated from the frozen machine-readable analysis artifact.

Supplementary Note: Dataset and shift construction

Common split structure

We study binary malware or network-attack classification using EMBER (Anderson and Roth, 2018), UNSW-NB15 (Moustafa and Slay, 2015), and ToN-IoT (Alsaedi et al., 2020). Every setting has a training set T, an in-distribution test set S_{ID}, and a shifted test set S_{OOD}. Classical and quantum kernels always use the same samples. Class balance is preserved in the constructed splits, so the measured change is not driven by a prevalence shift.

EMBER label-conditional tail shifts

The EMBER analysis uses the histogram and byte-entropy feature blocks and two reproducible stress tests. Both select OOD samples within each class. They are therefore described as *label-conditional tail shifts*, not as certified covariate shifts: conditioning the split on the class does not establish that $P(Y \mid X)$ is invariant.

In the $m1$ construction, each sample receives a sparsity score equal to the number of non-zero entries in the selected feature block. The OOD set contains the low- and high-score extremes within each class. The remaining samples form an ID pool from which training and ID test sets are sampled stratifiably.

In the $m2$ construction, the final training set is fixed first. Class-conditional centroids are computed in a MaxAbs-scaled, TruncatedSVD space fitted on training data only. Every non-training sample is scored by its Euclidean distance to its own-class centroid. The low- and high-distance extremes within each class form the exact-size OOD set, and the ID test set is drawn from the remainder. This construction therefore probes unusual geometry relative to the final training set.

Network-flow scenarios

The network experiments define UNSW-NB15 DoS, UNSW-NB15 Reconnaissance, and ToN-IoT Scanning scenarios. Numeric-only preparation removes identifiers, label-derived columns, and categorical protocol fields, retaining 39 UNSW-NB15 features and 89 ToN-IoT features. The largest observed absolute feature-label correlation is 0.62, attained by a legitimate traffic feature rather than a label-derived field; the public artifact includes the leakage audit.

Each scenario has a reference pool and a current pool. Two mechanisms are applied. The *m2-centroid* mechanism is the network analogue of EMBER $m2$: class-conditional centroids are fitted in a train-only MaxAbs+TruncatedSVD space and within-class distance tails define the OOD pool. The *attack-campaign regime shift* uses no geometric selection. Training and ID samples come from benign traffic plus all non-target attack categories, whereas OOD samples come from benign traffic plus the held-out target campaign. For UNSW-NB15, the pools also use different official capture partitions. This is consequently an unseen-campaign plus capture-partition shift, not a timestamp-defined temporal split.

Supplementary Note: Controlled kernel-swap protocol

Preprocessing and classifiers

All models use MaxAbs scaling followed by TruncatedSVD with

$$d \in \{4, 6, 8, 10, 12\}.$$

For fidelity kernels, each reduced coordinate is mapped linearly to $[0, \pi]$ before the angle-scale factor is applied. Preprocessing is fitted using the training split and shared across families.

Two learners consume precomputed Gram matrices. The primary learner is support-vector classification (SVC). The secondary learner is binary Gaussian-process classification using the standard Laplace approximation with a logistic link (Rasmussen and Williams, 2006). The GP classifier supplies approximate probabilities for the uncertainty analysis; no post-hoc calibration is applied. Comparisons are always made within a classifier.

Kernel candidates

The customary classical reference contains a cosine-normalized linear kernel and the radial-basis-function (RBF) kernel

$$K_{\mathrm{RBF}}(x, x') = \exp[-\gamma \|x - x'\|_2^2].$$

The extended classical family adds cosine-normalized polynomial kernels of degrees 2 and 3, the Laplacian kernel, and Matérn kernels with $\nu \in \{3/2, 5/2\}$. Base length scales are set by the median pairwise training distance. RBF, Laplacian, and Matérn candidates sweep factors

$$f \in \{0.1, 0.3, 1, 3, 10\}$$

relative to their base scale.

The quantum family uses fidelity kernels

$$K_Q(x, x') = \left|\langle \phi_Q(x) \mid \phi_Q(x') \rangle\right|^2$$

from four commonly used feature-map configurations: ZZ maps with one and two repetitions, a PauliXZ map with one repetition, and a Z map. Each is evaluated at angle scales $\{0.5, 1, 2\}$. These candidates cover representative low-qubit fidelity constructions rather than the full quantum-kernel design space.

Supplementary Table 1: Complete experimental protocol.

Component	Specification
Scenarios	EMBER; UNSW-NB15 DoS; UNSW-NB15 Reconnaissance; ToN-IoT Scanning
Shift mechanisms	EMBER $m1$ and $m2$; network m2-centroid and held-out attack-campaign shift
Master seeds	42, 123, 999
Split seeds	42, 123, 999, 7, 2024
Model seeds	42, 123, 999
Runs	$5 \times 3 = 15$ per evaluated configuration
Sizes $(n_{\text{train}}, n_{\text{ID}}, n_{\text{OOD}})$	$(1000, 500, 500)$, $(2000, 1000, 1000)$, $(4000, 1800, 1800)$
Dimensions	$d \in \{4, 6, 8, 10, 12\}$
Classifiers	Precomputed-kernel SVC; precomputed-kernel Laplace GP classifier
Classical family	Linear, RBF, polynomial (2, 3), Laplacian, Matérn (3/2, 5/2)
Quantum family	ZZ (1 and 2 repetitions), PauliXZ, Z map; fidelity overlap
Scale freedom	Five classical length-scale factors; three quantum angle factors
Regularization	Per-configuration SVC C, five-fold training-only cross-validation
Primary selection	P1′ ID-validation; no OOD labels; equal candidate budgets
Inference	Fixed-case conditional intervals; dataset-equal summaries; no population p-value

Regularization, deployment selection, and candidate budgets

For each of the 23 classical and 12 quantum kernel geometries, SVC regularization is selected from

$$C \in \{0.01, 0.1, 1, 10, 100\}$$

by five-fold stratified cross-validation on the training Gram matrix. Ties are broken toward the smaller C. Because C is resolved within each configuration before the family comparison, it is not counted as another family-level candidate.

The primary P1′ deployment protocol divides the original ID test split deterministically and class-stratifiably into an ID-validation half and a disjoint ID-test half. Within a run and family, the configuration with the highest ID-validation balanced accuracy is deployed and evaluated on the untouched ID-test and OOD-test splits. No OOD label enters selection.

The quantum pool contains 60 candidates in full-coverage groups and 20 in reduced-coverage groups after combining feature map, angle scale, and dimension. The frozen primary analysis compares it with an equal-size sample from the extended classical pool. Five thousand subsamples and three sampling schemes characterize budget sensitivity. The full 115-candidate classical pool is retained as a secondary sensitivity. P2 uses OOD information from other realizations and P3 selects on the same OOD test labels; neither is a deployable primary analysis.

Repeated runs and fixed-case inference

Each setting is realized with five split seeds and three model seeds, giving fifteen pipeline realizations per evaluated configuration. The eight scenario-groups share source pools and are not exchangeable; in particular, some constructed EMBER OOD samples recur across master seeds. The analysis therefore treats scenario-groups as fixed case studies. Per-group quantum-minus-classical effects are means over the five split-seed clusters with conditional t intervals. BCa bootstrap and leave-one-split-out calculations are secondary diagnostics. Across groups, we report dataset-equal means and leave-one-dataset-out ranges, not a population p-value.

Supplementary Table 2: Frozen specification definitions and dataset-equal quantum-minus-classical OOD balanced-accuracy effects. “Oracle” selects and evaluates on the same OOD test labels; “ID-val” uses a disjoint in-distribution validation split. SVC regularization is fixed at $C = 1$ or selected by training-only cross-validation; this axis is not applicable to GPC.

ID	Selection	SVC C	Classical reference	Budget	SVC	GPC
S1	OOD oracle	Fixed	Linear/RBF	Native	+0.0313	+0.0340
S2	Legacy ID	Fixed	Linear/RBF	Native	+0.0475	+0.0418
S3	OOD oracle	Fixed	Extended	Native	+0.0058	+0.0145
S4	Legacy ID	Fixed	Extended	Native	+0.0014	+0.0122
S5	OOD oracle	Train CV	Linear/RBF	Native	+0.0034	+0.0125
S6	ID-val	Train CV	Linear/RBF	Native	−0.0041	+0.0059
S7	OOD oracle	Train CV	Extended	Native	−0.0054	+0.0012
S8	ID-val	Train CV	Extended	Native	−0.0038	−0.0029
S9	OOD oracle	Train CV	Extended	Equal	−0.0032	+0.0038
S10	ID-val	Train CV	Extended	Equal	−0.0056	−0.0009

Supplementary Note: Frozen specification curve

The specification curve aggregates only complete paired endpoints already present in the versioned legacy and v4 artifacts; it does not refit a model or alter the experiment grid. Specifications were ordered S1–S10 before aggregation and are not ranked by their observed effect. Within a specification, realizations are averaged within setting, settings within scenario-group, scenario-groups within source dataset, and finally the three source datasets with equal weight. The specification-curve figure in the main article displays every scenario-group as well as the dataset-equal mean.

The range of the dataset-equal mean is 0.0531 for SVC and 0.0447 for GPC. This spread is descriptive rather than a confidence interval: the specifications share data and candidates, and they are deliberately different estimands. Its purpose is to expose how much of the headline depends on label access, regularization, reference-family strength, and search budget.

Supplementary Note: External TableShift validation

Prospective freeze and acquisition

The complete external protocol was committed before TableShift data were loaded and before any external quantum or classical result was computed. The source implementation is pinned to TableShift commit `fca9429814703a07e3902d005d46563a207b7f0a` (Gardner et al., 2023). The fixed tasks were College Scorecard, diabetes readmission, and ACS income; no fallback task was used. Their published domain-split variables are institution type, admission source, and geographic region, respectively. These tasks were selected to span education, health, and socioeconomic-policy data, not because of quantum-kernel performance.

TableShift's split roles are retained without reinterpretation. The `train` split fits preprocessing, kernels, classifiers, and SVC regularization; `validation` selects the deployed kernel; and `id_test` and `ood_test` are reporting endpoints. The source benchmark's `ood_validation` split is never loaded into the analysis pipeline. College Scorecard was acquired from its public CC0 archive because the pinned TableShift downloader unconditionally invokes authenticated Kaggle access; only that redundant download call was bypassed, while the pinned parser, schema, and domain splitting were retained. Source-file SHA-256 hashes and the container image digest are stored in the acquisition audit.

Sampling and leakage gates

For each source split and seed, rows are ordered by the SHA-256 digest of the task, split, seed, and original row position. Sampling is therefore deterministic and label blind. The first stratum uses $(1000, 250, 250, 500)$ train, validation, ID-test, and OOD-test rows; the second uses $(2000, 500, 500, 1000)$, with the smaller sample nested in the larger one. Seeds are $\{42, 123, 999, 7, 2024\}$. Automated gates verify source-row disjointness across split roles, nesting across sizes, both classes in every sample, source-schema preservation, and at least 12 non-constant post-encoding features. All 30 task–size–seed units passed; the encoded dimensionality ranged from 209 to 954 features.

All transformations are fitted on `train` only. Constant training columns are removed; numeric variables receive median imputation; Boolean, categorical, and string variables receive most-frequent imputation and one-hot encoding with unknown categories ignored. The combined matrix is MaxAbs scaled and mapped by TruncatedSVD to $d \in \{4, 6, 8, 10, 12\}$ using random state 20260729. Each reduced representation is standardized and mapped to $[0, \pi]$ with training-fitted parameters.

Estimands and conditional uncertainty

The kernel and classifier pools are identical to the security comparison: 60 quantum and 115 extended-classical geometries, per-candidate train-CV regularization for SVC, and the frozen Laplace GP classifier. For the primary effect, the quantum candidate with highest validation balanced accuracy is evaluated once on `ood_test`. Twelve whole five-dimension classical kernel blocks are sampled from the 23 available blocks, producing an equal pool of 60; its validation winner is evaluated on `ood_test`. The classical endpoint is the expectation over 5,000 such block subsamples with fixed seed 20260729.

Two diagnostics are computed from the same complete grid. The equal-budget oracle selects and evaluates on `ood_test`. The weak-fixed oracle uses only linear and RBF references, fixes SVC at $C = 1$, and also selects on the reported OOD labels. Protocol contraction is

$$\Delta_{\text{weak,fixed,oracle}} - \Delta_{\text{honest,equal,tuned}}.$$

Positive contraction means that replacing the controlled procedure with the weak oracle makes the quantum-minus-classical effect more favourable.

Seeds are averaged within task and size; the two nested sizes are averaged within task; and tasks receive equal weight. Conditional intervals resample the five seed clusters 9,999 times within task, preserving the nested-size relationship. Leave-one-task-out ranges are reported for the task-equal mean. The tasks are fixed case studies and receive no population p-value.

Supplementary Note: Metrics and geometry descriptors

Balanced accuracy on S_{OOD} is the primary endpoint. Family effects are

$$\Delta_{\text{OOD}} = \text{BAcc}^{(Q)}_{\text{OOD}} - \text{BAcc}^{(C)}_{\text{OOD}},$$

with positive values favouring the quantum family. ID balanced accuracy and the ID-to-OOD drop are secondary. The GP analysis additionally records log loss, Brier score, expected calibration error with 15 equal-width bins (Guo et al., 2017), and predictive entropy.

For a positive-semidefinite training Gram matrix with eigenvalues λ_i and $p_i = \lambda_i / \sum_j \lambda_j$, effective rank is

$$r_{\text{eff}} = \exp\left(-\sum_i p_i \log p_i\right)$$

(Roy and Vetterli, 2007). Centered kernel-target alignment (KTA) for labels $y \in \{-1, +1\}^n$ is

$$\text{KTA}(K, y) = \frac{\langle K_c, yy^\top \rangle_F}{n \|K_c\|_F},$$

Supplementary Table 3: Descriptive EMBER oracle results under fixed $C = 1$ (18 correlated settings). Mean and median effects are quantum minus classical balanced accuracy.

Selection	Classifier	Classical reference	Q wins	Mean Δ	Median Δ
Best-by-OOD	SVC	linear+RBF	17/18	+0.0374	+0.0426
	SVC	extended	15/18	+0.0044	+0.0064
	GP	linear+RBF	15/18	+0.0283	+0.0339
	GP	extended	11/18	+0.0023	+0.0029
Best-by-ID	SVC	linear+RBF	18/18	+0.0775	+0.0794
	SVC	extended	18/18	+0.0149	+0.0152
	GP	linear+RBF	18/18	+0.0649	+0.0664
	GP	extended	18/18	+0.0192	+0.0166

where K_c is doubly centered (Cristianini et al., 2001; Cortes et al., 2012). KTA is computed separately on training, ID, and OOD blocks.

We also record the mean and standard deviation of off-diagonal entries, cross-split similarities, and the geometric difference

$$g(K_C \to K_Q) = \left\| \sqrt{K_Q}(K_C + \lambda n I)^{-1} \sqrt{K_Q} \right\|_\infty^{1/2}$$

(Huang et al., 2021). Geometry-performance association is assessed within scenario-groups across all kernel-dimension cells using Spearman correlation, with partial association controlling for family and length scale. Leave-one-dataset-out prediction tests portability, and nearest-effective-rank matching compares families within run and dimension. These are observational diagnostics, not causal estimates or deployable selectors.

Supplementary Note: Why the oracle comparison is optimistic

The initial evaluation selected the best candidate using the OOD test labels themselves and left SVC regularization at $C = 1$. Against linear and RBF kernels, this produces a mean EMBER OOD effect of +0.0374 for SVC and +0.0283 for the GP classifier (Table 3). Expanding the classical family nearly removes this apparent gain even before honest selection is imposed. Because the same target labels select and score the candidate, these values are descriptive upper bounds rather than deployable effect estimates.

Supplementary Note: Per-group geometry and rank matching

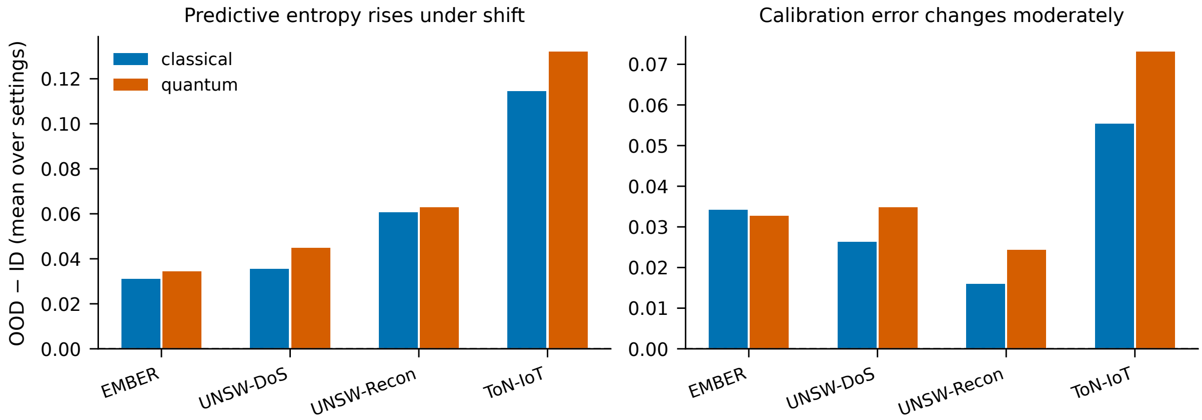
Table 4 supplies the values summarized graphically in the main article. Effective-rank association changes sign across regimes, whereas OOD KTA is more consistently positive. Nearest-rank matching favours the classical member in median accuracy in every group.

Supplementary Note: Predictive uncertainty under shift

The Laplace GP probabilities are approximate and are assessed rather than assumed to be calibrated. Predictive entropy rises from ID to OOD for both families in every scenario-group. Expected calibration error changes moderately and comparably (Figure 1). Because the honestly selected fidelity kernels have no accuracy advantage, the uncertainty analysis provides no evidence of a calibration benefit unavailable to the classical family. Entropy increase alone is not evidence of good calibration.

Supplementary Table 4: Geometry-OOD association and nearest-effective-rank matching for SVC. "Partial" controls for kernel family and length scale.

Scenario	ρ rank	Partial	ρ KTA	Matched Δ	Q better
EMBER $m1$	+0.32	+0.36	+0.30	−0.0094	33%
EMBER $m2$	+0.80	+0.82	+0.63	−0.0089	22%
UNSW-DoS (campaign)	+0.44	+0.41	+0.64	−0.0089	27%
UNSW-DoS (constructed)	−0.14	−0.08	+0.39	−0.0057	29%
UNSW-Recon (campaign)	+0.06	+0.06	+0.22	−0.0047	37%
UNSW-Recon (constructed)	−0.15	−0.08	+0.37	−0.0061	30%
ToN-IoT (campaign)	+0.25	+0.27	+0.46	−0.0060	24%
ToN-IoT (constructed)	+0.65	+0.63	+0.59	−0.0013	35%



Supplementary Figure 1: Mean ID-to-OOD change in predictive entropy (left) and expected calibration error (right) for the P1′-selected configuration of each family, aggregated by dataset.

Supplementary Note: Finite-shot fidelity-estimation model

For $s \in \{128, 512, 2048, 8192\}$ shots, every exact fidelity p is replaced by a binomial estimate $\hat{p} \sim \text{Binomial}(s, p)/s$. Square training blocks are symmetrized, clipped to $[0, 1]$, assigned an exact unit diagonal, and projected to the positive-semidefinite cone. Rectangular test blocks receive sampling and clipping only. The procedure isolates fidelity-estimation noise; it does not model a specific device, circuit compilation, gate noise, readout error, or mitigation.

The main article reports the aggregate perturbation results. Two qualifications matter. First, PSD projection changes the spectral object whose effective rank is measured, so both pre- and post-projection deviations are retained in the artifact. Second, low selected-configuration agreement does not imply a comparable accuracy penalty: the family optimum is flat enough that selection regret remains negligible even when the candidate identity changes.

Supplementary Note: Machine-readable result map

The complete per-setting results are intentionally not reproduced as static tables. They include all eight scenario-groups, three sizes, master and split seeds, both classifiers, protocols P1′, P2, and P3, budget-resampling draws, geometry descriptors, finite-shot perturbations, and frozen input hashes. The canonical entry point is

```
python scripts/reproduce_v4.py --stage all
```

and the generated tables and figures used for the manuscript are under `results/v4/tables/` and `manuscript/`. The archived artifact is available at `https://doi.org/10.5281/zenodo.`

19147649; the development repository is `https://github.com/roberto-fernandez-barrios/kernel_shift_framework`.

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